

AI Supervisor Assistant

Project Documentation

- React + Vite
- Node.js + Express
- Supabase
- Groq AI
- Resend



- PLATFORM

Vercel + Render
- DATABASE

Supabase PostgreSQL
- AI MODEL

Groq llama-3.3-70b

1. Project Overview

AI Supervisor Assistant is a full-stack project management platform that uses artificial intelligence to help managers create projects, generate tasks and staffing roles, hire team members, track progress, and communicate — all in one place.

The platform supports two roles: **Manager** (full access — create projects, hire, manage) and **Team Member** (view assigned projects, update task progress, use chat). Language can be switched between English, Spanish, French, and German at any time.

2. Technology Stack

Frontend

Technology	Version	Purpose
React	18	UI framework
Vite	5	Build tool and dev server
Tailwind CSS	4	Styling
React Router	v6	Client-side routing
i18next / react-i18next	Latest	Translations (EN/ES/FR/DE)
Recharts	Latest	Donut chart / data visualization
Lucide React	Latest	Icon library
React Markdown	Latest	Markdown in AI chat

Backend

Technology	Purpose
Node.js + Express	REST API server (port 5000)
Supabase (PostgreSQL)	Database, Auth, and File Storage
LangChain + Groq (llama-3.3-70b)	AI task/role generation and project chat
Google Generative AI Embeddings	Vector embeddings for semantic task search
Resend	Transactional email delivery
node-cron	Weekly report email scheduling

Deployment

Service	Platform	URL
Frontend	Vercel	https://ai-supervisor-app.vercel.app
Backend	Render	https://ai-supervisor-app.onrender.com
Database / Auth / Storage	Supabase	Managed PostgreSQL

3. Project Structure

Repository Layout

```
ai-supervisor-assistant/  
  Frontend/src/  
    i18n.js – translations config (EN/ES/FR/DE)  
    main.jsx – app entry point  
    App.jsx – route definitions  
    Context/ – Auth + AppSettings providers  
    Pages/ – Dashboard, AddProject, Candidates, Settings  
    Components/ – Sidebar, FloatingChat, Dashboard/, Project/, Settings/, Candidate/  
  backend/src/  
    index.js – Express entry point (port 5000)  
    agents/parseProject.js – AI generation logic  
    Controllers/ – auth, user, project, task, members, calendar, uploads, chat  
    Routes/ – REST route definitions  
    Middleware/ – auth guard, role check  
    lib/ – aiConfig.js, supabaseClient.js  
    Utils/ – email templates, NotificationService, WeeklyReportJob
```

4. Environment Variables

Backend (.env)

Variable	Description
PORT	Server port (default 5000)
CLIENT_URL	Frontend URL for CORS and email links
SUPABASE_URL	Supabase project URL
SUPABASE_ANON_KEY	Supabase anonymous key
SUPABASE_SERVICE_ROLE_KEY	Supabase service role key (bypasses RLS)
GROQ_API_KEY	Groq API key for LLM (llama-3.3-70b)
GOOGLE_API_KEY	Google AI key for embeddings
RESEND_API_KEY	Resend API key for email delivery

Frontend (.env)

Variable	Description
VITE_API_URL	Backend base URL (e.g. https://ai-supervisor-app.onrender.com)
VITE_SUPABASE_URL	Supabase project URL
VITE_SUPABASE_ANON_KEY	Supabase anonymous key

4. Features

Manager Features

Project Creation — Upload a PDF or enter a description — AI generates tasks and roles automatically using Groq Llama-3.3-70b.

Project Update — Upload a new PDF — AI merges new requirements with existing tasks without overwriting them.

AI Role Assignment — AI assigns role_id to tasks automatically when opening the Candidates page.

Candidate Scoring — Team members are scored by skill match % against 15 expected skills per role type.

Hiring & Firing — Hire candidates into roles with one click. Sends a hiring email via Resend.

Task Management — Edit task what/how fields, update status, assign to hired team members.

Calendar Events — Create meetings or deadlines with time slots, invite project members, receive emails.

Email Alerts — Instant emails on task updates, project changes, and calendar events.

Weekly Reports — Automated Monday 6AM email summarizing all project progress per user timezone.

Team Chat — AI assistant chat per project + DMs and group chats with team members.

Team Member Features

Project View — See only projects they are hired into.

Task Progress — Update status (Not Started, In Progress, Delayed, Complete) for their role tasks only.

Inline Editing — Edit the What and How fields of tasks assigned to their role.

Chat — AI assistant + DMs + group chats with any project coworker.

Settings — Profile, notifications, language and date format preferences.

Dashboard — Personal stats — Active Projects, Total Tasks, Completed Tasks, Overdue Tasks.

5. Installation & Running Locally

Prerequisites

- Node.js v18+
- npm v9+
- Supabase project
- Groq API key
- Google AI API key
- Resend API key

Install Dependencies

```
# Backend cd backend npm install npm install node-cron # Frontend cd Frontend npm  
install npm install i18next react-i18next react-markdown
```

Run Locally

```
# Terminal 1 – Backend cd backend && npm run dev # Starts on http://localhost:5000 #  
Terminal 2 – Frontend cd Frontend && npm run dev # Starts on http://localhost:5173
```

6. Database Tables

Table	Description	Key Columns
profiles	User accounts and preferences	id, role, email_alerts, weekly_reports, preferred_language
projects	Projects with JSONB roles array	id, owner_id, name, status, roles (jsonb), deadline
tasks	Project tasks with AI assignment	id, project_id, title, what, how, status, role_id, role_title
project_members	Hired team members per role	project_id, user_id, role_id, role_title, hired_at
uploads	PDF uploads and extracted text	id, project_id, filename, file_url, raw_text
calendar_events	Meetings and deadlines	id, type, event_date, event_time, participants (uuid[])
chat_messages	AI and team chat messages	id, project_id, channel, sender_id, role, content
chat_channels	DM and group channels	id, project_id, name, members (uuid[]), is_group

7. Email Notifications

Trigger	Recipients	Respects Alerts Toggle
Task status updated	Manager + hired members with email_alerts ON	Yes
Project updated (new tasks/roles)	Hired members with email_alerts ON	Yes
Deadline event created	All project members	No — always sent
Meeting event created	All invited participants	No — always sent
Candidate hired	Hired candidate	No — always sent
Weekly report	Users with weekly_reports ON	Scheduled Monday 6AM